ModelIt!

Teacher's Guide

Why You Can't Slide Forever

The Hidden Energy Transfers That Stop Everything

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-PS3-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS3-5

Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students construct and present arguments using model evidence that energy is transferred when kinetic energy changes.
Disciplinary Core Idea	**PS3.B Conservation of Energy and Energy Transfer** — When the motion energy of an object changes, there is inevitably some other change in energy at the same time — energy is conserved.
Crosscutting Concept	**Energy and Matter** — Students track energy as it transfers from kinetic to thermal form, demonstrating that energy is conserved even as it changes form.

CAST Testing Alignment

MULTIPLE CHOICE:

A toy car rolls down a ramp and across a rough carpet, gradually slowing to a stop. Which statement best explains where the car's kinetic energy went?

CONSTRUCTED RESPONSE:

A student claims that when a ball stops rolling, the energy has been 'used up' and no longer exists. Using evidence from your model, construct an argument explaining what actually happens to the energy.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Energy is one of the most fundamental concepts in all of science. The law of conservation of energy states that energy cannot be created or destroyed — only transferred or transformed. Understanding energy transfer is key to everything from designing brakes to building power plants.

Kinetic Energy

Kinetic energy is the energy of motion, calculated as $KE = \frac{1}{2}mv^2$. This means kinetic energy depends on both mass and speed — but speed has a SQUARED effect. Doubling the mass doubles the energy, but doubling the speed QUADRUPLES it. This is why high-speed crashes are so much more destructive.

Friction and Heat

Friction is a force that opposes motion between surfaces in contact. It converts kinetic energy into thermal energy (heat) through microscopic interactions between surface irregularities. This is why brakes get hot, why rubbing your hands warms them, and why meteors burn up entering the atmosphere.

Conservation of Energy

The total energy in a closed system remains constant. When a sliding object stops, its kinetic energy hasn't disappeared — it has been transformed into thermal energy (heat in the surfaces and surrounding air). If you could measure all the heat produced, it would exactly equal the kinetic energy lost.

Real-World Applications

Engineers use energy transfer principles everywhere: car brakes convert kinetic to thermal energy (disc brakes can reach 300°C), regenerative braking in electric cars converts kinetic back to electrical energy, and heat shields on spacecraft manage the enormous thermal energy from atmospheric friction.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify initial speed, surface friction, kinetic energy, and thermal energy output as components of the energy transfer system.

E **Establish Relationships**
Students determine that friction negatively affects kinetic energy but positively transfers it to thermal energy — energy is conserved through transformation.

V **Visualize & Model**
Students build a model showing how kinetic energy is converted to thermal energy through friction.

E **Evaluate Outcomes**
Students run ice, concrete, and carpet scenarios to compare energy transfer rates on different surfaces.

R **Revise & Extend**
Students add air resistance or mass to explore more complex energy transfer dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You push a ball. It rolls. Then it stops. WHERE DID THE ENERGY GO?"*

Watch For

Roll a ball across the floor. When it stops, ask: The energy didn't vanish. Where is it?

Slide 2: Learning Objectives

3 min

Say: *"Today we're solving one of physics' most important puzzles — what happens to energy."*

Watch For

Have students read objectives. Pre-teach 'kinetic energy' and 'friction.'

Slide 3: Big Question

3 min

Say: *"In space, things move forever. On Earth, everything stops. Why the difference?"*

Watch For

Discussion: What's different about space vs. Earth? (No friction in space.)

Slide 4: LEVER Framework

2 min

Say: *"We'll model where energy goes when motion stops."*

Watch For

Preview LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What starts with energy? What takes energy away? Where does it end up?"*

Watch For

Guide sorting. Discuss why speed and surface are inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"Friction TAKES kinetic energy. But it doesn't destroy it — it TRANSFORMS it."*

Watch For

Have students rub hands to feel KE→thermal. Build relationships.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test: ice vs. concrete vs. carpet. Which steals energy fastest?"*

Watch For

Students predict first. Compare surface scenarios.

Slide 8: Discoveries

5 min

Say: *"Energy is NEVER destroyed. So where is it right now?"*

Watch For

Touch the floor where the ball stopped. Is it slightly warmer?

Slide 9: STEM Challenge

12 min

Say: *"Engineer a track that keeps a toy car moving as far as possible!"*

Watch For

Groups design tracks testing different surfaces for minimum friction.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Initial Speed, Surface Friction
- **INTERNAL:** Kinetic Energy, Thermal Energy Output

Relationships

- **Initial Speed to Kinetic Energy:** POSITIVE (+) — Faster initial speed means more kinetic energy. Speed has a squared relationship with KE — doubling speed quadruples kinetic energy.
- **Surface Friction to Kinetic Energy:** NEGATIVE (-) — More friction means kinetic energy is lost faster. Rough surfaces oppose motion more strongly, slowing objects down quickly.
- **Surface Friction to Thermal Energy Output:** POSITIVE (+) — More friction converts more kinetic energy into heat. The energy isn't lost — it's transformed into thermal energy in the surfaces.

Key Discoveries

- Energy is never created or destroyed — it only changes form (conservation of energy)
- Friction converts kinetic energy into thermal energy — that's why your hands get warm when you rub them
- Smoother surfaces mean less friction and slower energy conversion — objects slide farther
- Kinetic energy depends on speed SQUARED — doubling speed quadruples the energy

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Energy disappears when objects stop"	Energy is NEVER created or destroyed — this is the law of conservation of energy. When a ball stops rolling, its kinetic energy has been converted to thermal energy (heat) in the surface and surrounding air.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Friction only slows things down"	Friction also enables motion! Without friction, you couldn't walk (feet would slip), cars couldn't drive (tires wouldn't grip), and you couldn't hold anything. Friction is essential for controlled movement.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavier objects always stop faster"	Heavier objects have MORE kinetic energy at the same speed (KE = $\frac{1}{2}mv^2$), so they actually require MORE friction force to stop. This is why loaded trucks need longer stopping distances than empty ones.